Part D: Best model for recommendation to StadioEquities (client)

1. Executive Summary

Three machine learning models Logistic Regression (LR), Random Forest (RF), and XGBoost (XGB) were trained and evaluated on the UCI Bank Marketing dataset to predict whether a bank client will subscribe to a term deposit. The dataset comprises 45,211 phone-campaign records from a Portuguese bank, with approximately 11% positive responses, making this a highly imbalanced binary classification problem. The data-leakage column "duration" (call length, only known after the call ends) was removed. Evaluation used 5-fold cross-validation repeated 10 times (50 runs) with 95% confidence intervals and pairwise statistical testing (paired t-test and Mann–Whitney U).

Key Finding: XGBoost is the recommended model, achieving the highest F1 score for the minority class (0.459 on the test set, 0.446 in cross-validation) and statistically significant superiority over both alternatives ($p < 0.001$).

2. Model Comparison Results

Table 1: Test-set performance (9,043 held-out samples)

Model	Accuracy	F1 (yes)	Precision	Recall	Key Observation
Logistic Regression	0.886	0.373	0.27	0.62	Highest recall; many false positives
Random Forest	0.893	0.318	0.46	0.21	High accuracy but misses 79% of subscribers
XGBoost	0.891	0.459	0.38	0.59	Best precision–recall balance

Table 2: Cross-validation results (5-fold × 10 repeats, 95% CI via t-distribution)

Model	Mean F1	Std Dev	95% CI Lower	95% CI Upper
Logistic Regression	0.3731	0.0108	0.3701	0.3761
Random Forest	0.3137	0.0154	0.3093	0.3181
XGBoost	0.4458	0.0101	0.4430	0.4487

Statistical Significance: All pairwise comparisons are highly significant ($p < 0.001$) under both the paired t-test and the Mann–Whitney U test, confirming XGBoost is statistically superior to both LR and RF.

3. Recommendation: XGBoost

XGBoost is recommended based on three dimensions: highest F1 score (0.459 test / 0.446 CV), best precision–recall balance (59% recall, 38% precision), and statistically significant superiority with tight confidence intervals indicating stable generalisation.

4. Improvement Strategies

Table 3: Recommended strategies to further improve XGBoost performance

Strategy	Description	Expected Impact
Engineered Features	Switch from raw to the 8 engineered features (age_group,	Exploits non-linear relationships and domain interactions

	balance_group, was_contacted_before, total_contacts, prior_success, season, has_both_loans, balance_per_contact)	
Hyperparameter Optimisation	GridSearchCV, RandomizedSearchCV, or Bayesian optimisation (Optuna) on n_estimators, learning_rate, max_depth, subsample, colsample_bytree, reg_alpha/lambda	Identifies optimal configuration beyond manual defaults
Decision Threshold Tuning	Lower threshold from 0.5 to 0.3–0.4; use precision-recall curve to align with business objective	Increases recall for minority class; balances trade-off
Advanced Resampling	Apply SMOTE or ADASYN within the CV pipeline to generate diverse synthetic positive examples	Gives model more representative minority instances
Feature Selection	Recursive Feature Elimination (RFE), importance ranking, or L1 regularisation to remove noise from 50 encoded features	Reduces overfitting; improves generalisation
Ensemble Methods	Stacking ensemble or soft voting combining XGBoost (high precision) with LR (high recall)	Leverages complementary strengths for higher F1

6. Adaptation of selected models to StadioEquities (client)

The XGBoost model can be reused, but it requires significant adaptation for STADIOEquities. Unlike the UCI Bank Marketing dataset, STADIOEquities operates in a digital investment environment with 2.3 million accounts and rich behavioural data. The prediction target should shift to **first deposit after registration**, requiring management of severe class imbalance and evaluation using **precision-recall AUC**. Features must be rebuilt around customer behaviour and financial activity, including onboarding progress, engagement, trading patterns, portfolio metrics, balances, and support interactions, while incorporating text analytics from support tickets and app reviews. Longitudinal data must be transformed into point-in-time features with strict leakage controls. Decision thresholds should vary by use case, such as activation campaigns versus premium-product targeting. While XGBoost's strengths remain applicable, successful deployment depends primarily on establishing a unified client-data platform that integrates behavioural, financial, and service data and supports large-scale model training.

5. Conclusion

XGBoost is the recommended model for predicting term deposit subscriptions, demonstrating statistically significant superiority over both Logistic Regression and Random Forest. With the proposed improvements adoption of engineered features, systematic hyperparameter optimisation, and decision threshold tuning the model’s F1 performance can be expected to improve further, leading to more effective targeting of potential subscribers and a higher return on campaign investment.